



ST BENEDICT'S

SUBJECT	Physical Science Paper 1	DATE	22 July 2019
GRADE	11	MARKS	165
EXAMINER	Mr McCoy	MODERATOR	Mr Ntombela
NAME		DURATION	150 minutes
CLASS			
LOW ORDER	50%	HIGH ORDER	50%

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 14 pages, an Answer Sheet of 2 pages and a Data Sheet of 2 pages. Please make sure that your question paper is complete. (You may remove the Data sheet)
2. Answer ALL the questions.
3. Read the questions carefully.
4. Use the data and formulae whenever necessary.
5. **Start each question on a new page.**
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
8. Where appropriate express answers to **TWO** decimal places.
9. It is in your own interest to write legibly and to present your work neatly.

Question	1	2	3	4	5	6	7	Total
Marks	16	34	31	20	24	14	26	165

QUESTION 1- MULTIPLE CHOICE

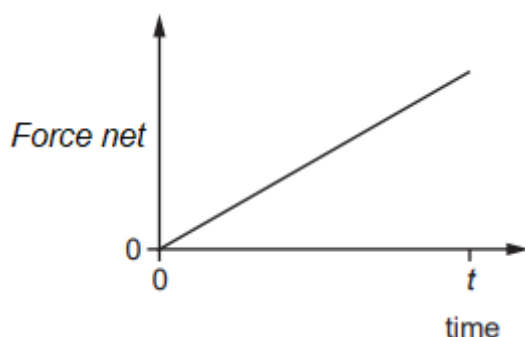
16 MARKS

Answer these questions on the multiple-choice Answer Sheet. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

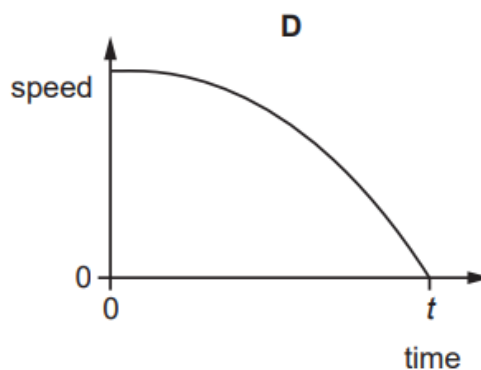
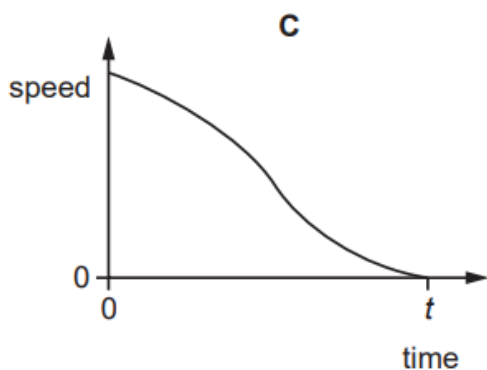
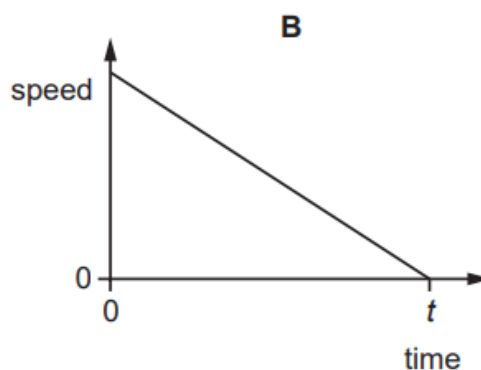
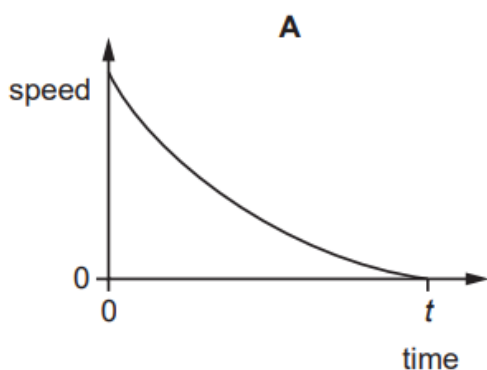
1.1 Which of the following pairs of physical quantities contains ONLY vector quantities? (2)

- A velocity, energy
- B acceleration, distance
- C work, speed
- D displacement, velocity

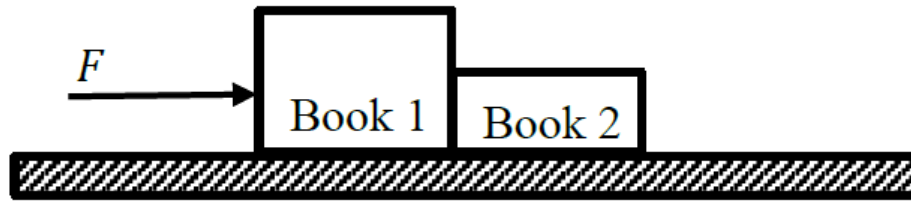
1.2 A driver stops his car in time, t , by gradually increasing the total braking force on the car. The graph shows the resultant force on the car.



Which graph shows how the **speed** of the car will vary during this time? (2)



- 1.3 Peter pushes two books on a frictionless surface with a force F as shown in the diagram.



The force that Book 1 exerts on Book 2 is X . The force that Book 2 exerts on Book 1 is Y . The magnitude of force X compared to force Y is:

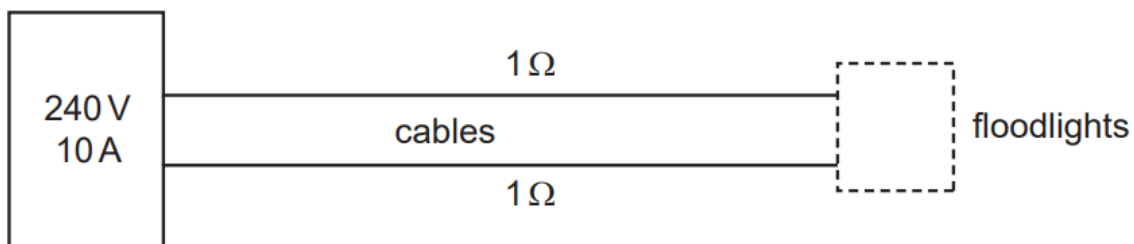
- A $X = Y$
B $X > Y$
C $X < Y$
D Depends on the acceleration of the system (2)
- 1.4 Two forces, 4 N and 11 N, simultaneously act on a point. Which of the following **cannot** be the magnitude of their resultant?
A 5 N
B 7 N
C 9 N
D 15 N (2)
- 1.5 An object has a weight of 88,20 N on the Earth. The gravitational field strength on the Moon is $1,64 \text{ m}\cdot\text{s}^{-2}$. What are the weight and the mass of the object when on the Moon?

	Weight (N)	Mass (kg)
A	14,76	1,5
B	14,76	9,0
C	88,20	1,5
D	88,20	9,0

(2)

- 1.6 The diagram shows a portable generator connected by cables to floodlights. The generator produces a current of 10 A at a constant potential difference (p.d.) of 240 V.
The total resistance of the cables is $2\ \Omega$.

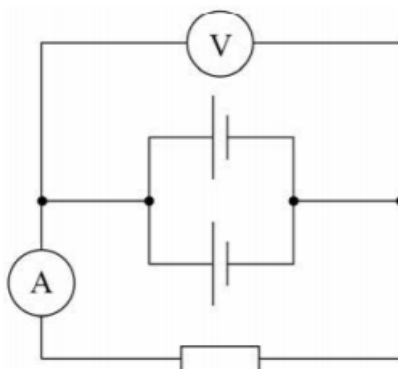
generator



What is the p.d., V , across, and the power P delivered to, the floodlights? (2)

	Voltage (V)	Power (W)
A	220	2400
B	220	2200
C	230	2400
D	230	2300

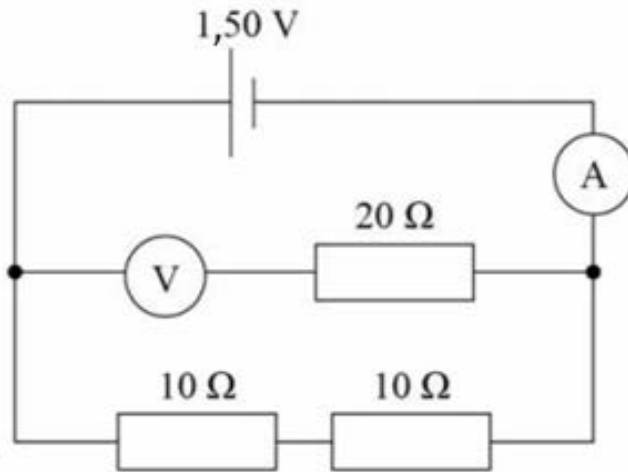
- 1.7 A circuit of two identical cells, a resistor, an ammeter and a voltmeter. The cells each have an emf of 3,0 V and the resistor has a resistance of $12\ \Omega$.
The cells have negligible internal resistance.



Which row shows the reading on the voltmeter and the ammeter? (2)

	Voltage (V)	Current (I)
A	3,0	0,25
B	3,0	0,50
C	6,0	0,25
D	6,0	0,50

- 1.8 The circuit shows a cell with negligible internal resistance connected in a circuit with three resistors, an ammeter and a voltmeter.



Which row shows the readings on the ammeter and the voltmeter?

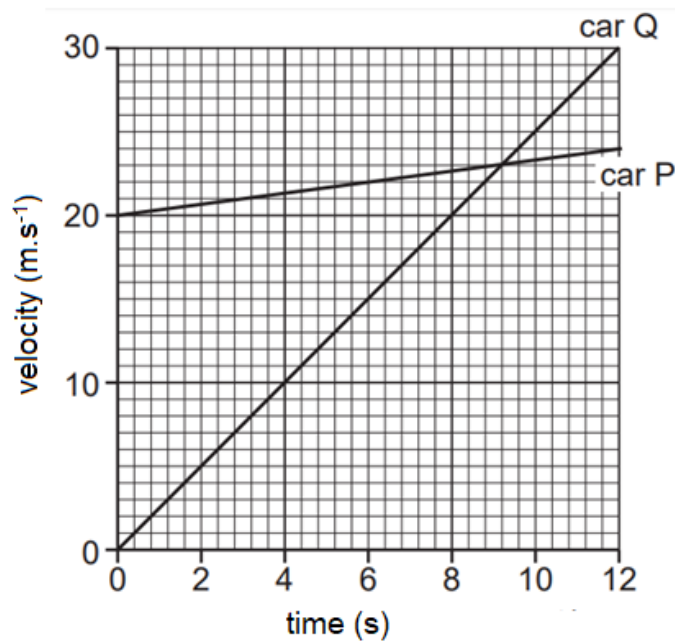
(2)

	Current (I)	Voltage (V)
A	0,075	0,75
B	0,075	1,50
C	0,150	0,75
D	0,150	1,50

QUESTION 2

34 MARKS

2.1 The variation with time of the velocity of two cars P and Q is shown below.

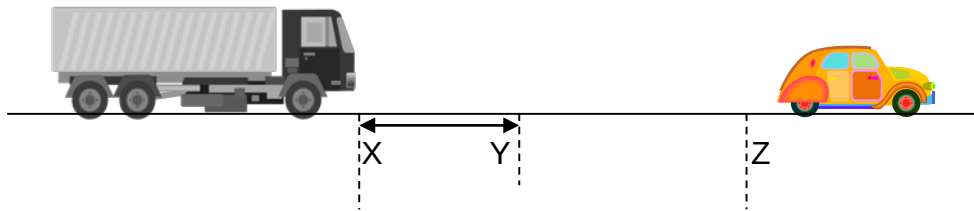


The cars travel in the same direction along a straight road. Car P passes car Q at time $t = 0$.

- 2.1.1 Define *velocity*. (2)
- 2.1.2 The speed limit for cars on the road is 100 km.h^{-1} . State and explain whether car Q exceeds the speed limit. (3)
- 2.1.3 Calculate the magnitude of the acceleration of car P (3)
- 2.1.4 Determine the distance between the two cars at time $t = 12 \text{ s}$. (4)
- 2.1.5 From time $t = 12 \text{ s}$, the velocity of each car remains constant at its value at $t = 12 \text{ s}$. Determine the time t at which car Q passes car P. (4)

2.2 A truck is moving forward at a constant speed of $20 \text{ m}\cdot\text{s}^{-1}$ on a horizontal road.

At point X the driver sees a stationary car ahead and hits the breaks $0,35 \text{ s}$ later at Y. The magnitude of the average acceleration of the truck is then $2,5 \text{ m}\cdot\text{s}^{-2}$. The truck manages to stop 2 m away from the car at point Z.



2.2.1 What is the reaction time of the driver? (1)

2.2.2 Calculate how far the truck travels before the brakes are applied. (2)

2.2.3 Calculate how long it takes before the truck comes to rest from the **moment** the car is seen. (4)

2.2.4 Calculate how far point X is from the car. (4)

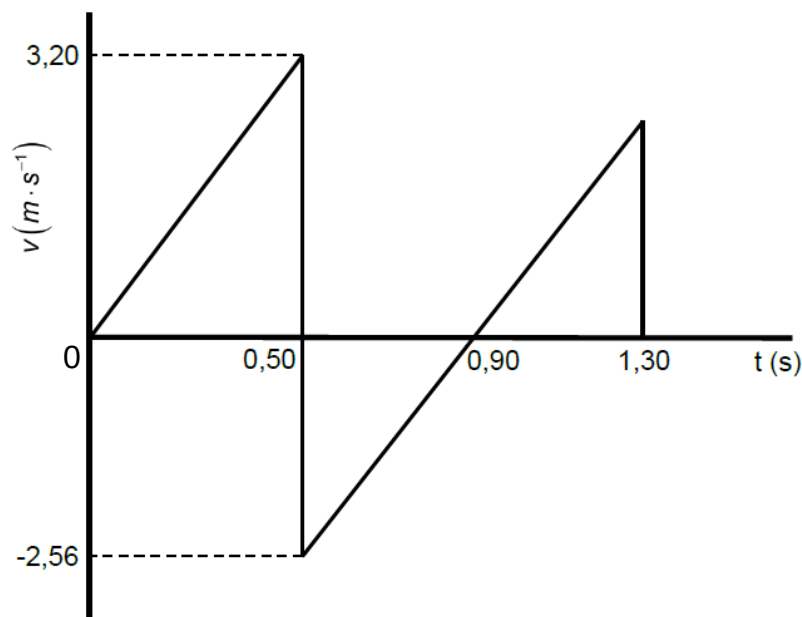
2.2.5 Draw a v/t sketch graph for the motion of the truck from X to Z. (3)

2.3 A car travels a distance **R** in a time **t** accelerating uniformly from rest. What distance, **in terms of R**, will the car travel if it accelerates for twice the time? Assume rectilinear motion. (4)

QUESTION 3

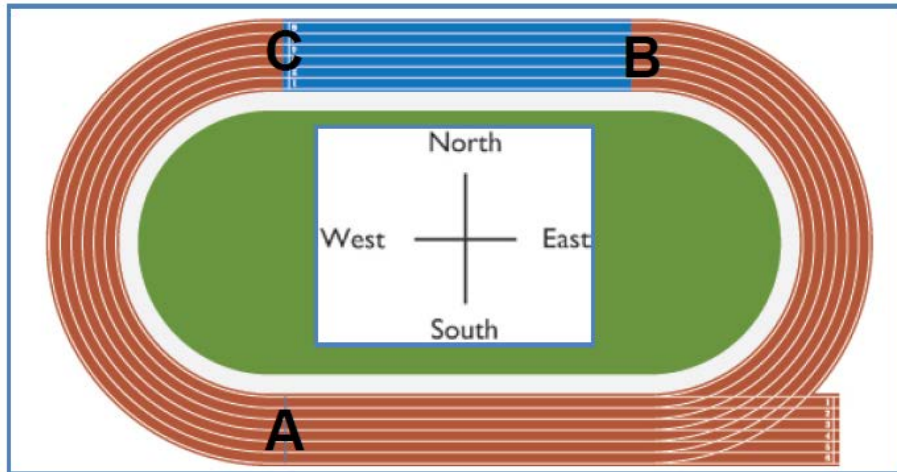
31 MARKS

- 3.1 A ball on a distant planet is dropped from a certain height. The velocity-time graph represents the motion of the ball as it bounces vertically on a solid floor. The time of the bounce is negligible. The planet has no atmosphere so there is zero air resistance.



- 3.1.1 Define *acceleration*. (2)
- 3.1.2 Describe the motion of the ball for 0,5 – 0,9 seconds. (4)
- 3.1.3 Calculate the height from which the ball was dropped. (3)
- 3.1.4 Draw a labelled free-body diagram showing the forces acting on the ball at 0,7 s. Ignore air resistance. (2)
- 3.1.5 Was any energy lost during the bounce? Briefly explain your answer. (2)
- 3.1.6 State the magnitude of the velocity with which the ball hits the ground at 1,30 s. (2)
- 3.1.7 Sketch an acceleration versus time graph of the ball from 0 s to 0,9 s. (3)
- 3.1.8 On the set of axes **on the answer sheet**, sketch the corresponding position-time graph for the motion of the ball from 0 s to 1,3 s. Use the point of release as position zero. Maintain the direction used in the graph above. (4)

- 3.2 An athlete runs in an anticlockwise direction around an 800 m long track shown below. The athlete starts from point A, running east.



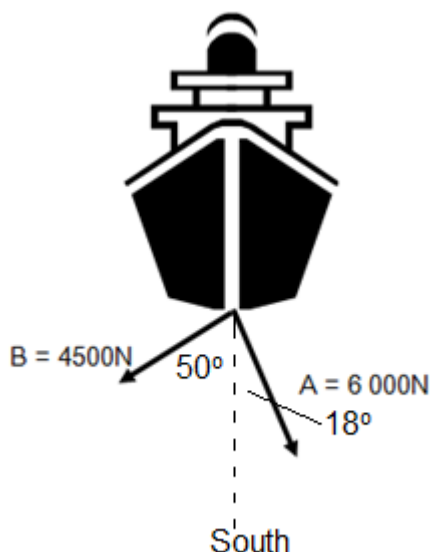
[Source: <<https://cdn.tutsplus.com/vector>>]

- 3.2.1 It takes the athlete 1 minute and 45 seconds to run half way around the track to point B. Calculate the average speed of the athlete. (3)
- 3.2.2 Suppose the athlete continues running at a constant speed, what is the athlete's **instantaneous velocity** at point C? (2)
- 3.2.3 Use the point A – C to explain the difference between distance and displacement. (4)
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QUESTION 4

20 MARKS

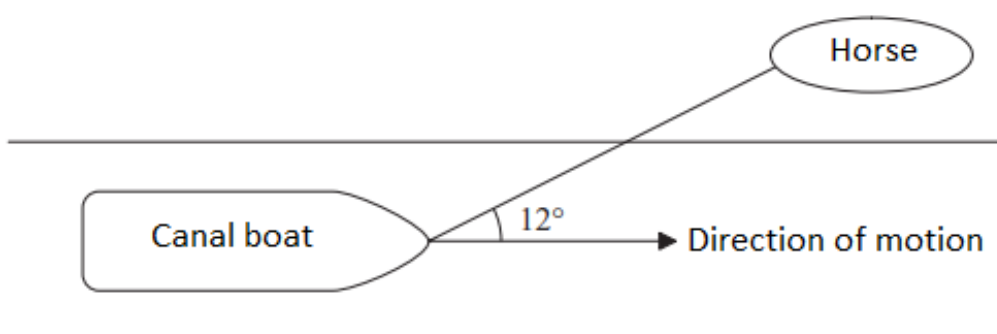
4.1 A ship is being towed by two tug-boats with forces A and B, as in the picture.



4.1.1 Define *resultant force*. (2)

4.1.2 Determine the resultant force on the ship. Give your answer in bearing. (6)

4.2 A horse is pulling a canal boat, at constant velocity, using a rope at 12° to the direction of motion of the boat. The tension in the rope is 1150 N. The diagram below shows the boat viewed from above.



4.2.1 Define *speed*. (2)

4.2.2 The canal boat is moving at a steady speed. Calculate the resistive (frictional) force opposing the boat's forward motion. (3)

4.2.3 Why is it necessary to have someone steer the boat in order to keep it in the direction of motion? (2)

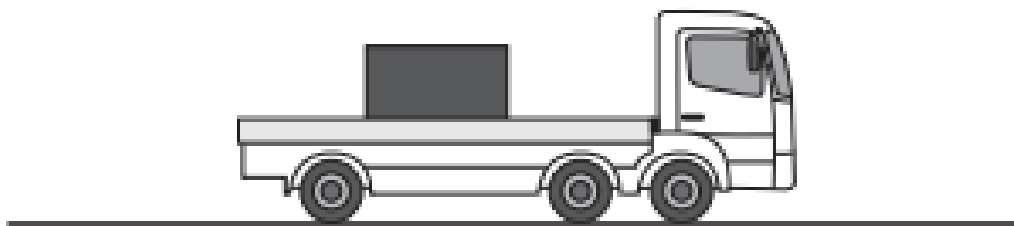
4.2.4 Draw a vector addition diagram of all the forces acting on the boat in the same plane at the tensional force. (3)

4.2.5 Explain why using a longer rope could allow the horse to move the boat at the same velocity, but with a lower tension in the rope. (2)

QUESTION 5

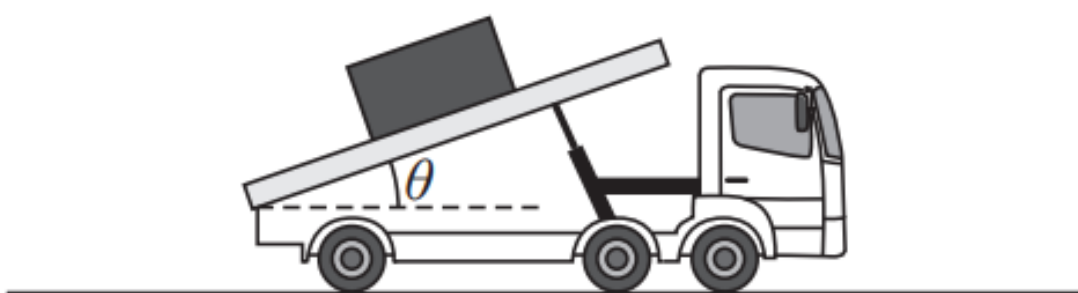
24 MARKS

There is a box of mass 200 kg on the back of a truck. The box is not tied to the truck.



- 5.1 State *Newton's first law*. (2)
- 5.2 Define *friction*. (2)
- 5.3 The maximum frictional force between the truck and the box is 570 N.
- 5.3.1 Refer to Newton's first law to explain why this limits the maximum acceleration that the truck can have without the box falling off. (4)
- 5.3.2 Will the kinetic friction between the truck and the box be GREATER THAN, LESS THAN or EQUAL to 570 N. Explain your answer. (3)

The truck has reached its destination, with the box still on. The back of the truck is tilted at an angle θ to the horizontal.



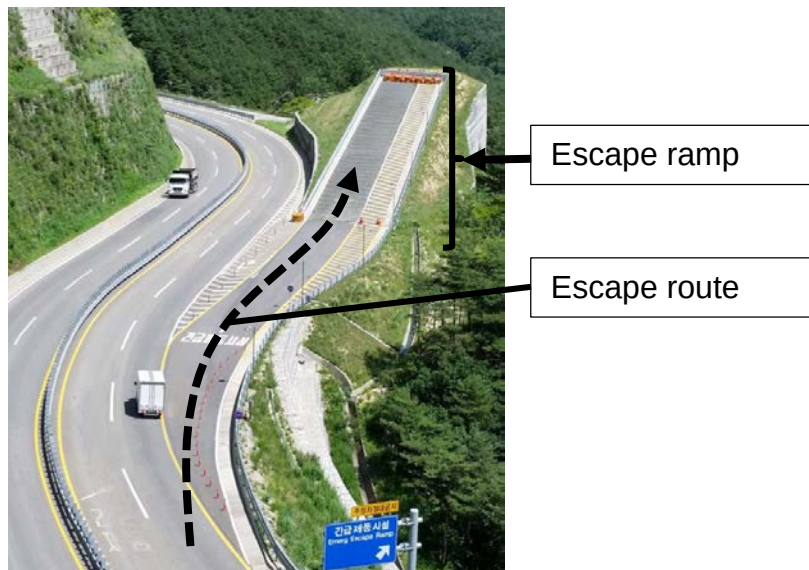
- 5.4 Draw a labelled free-body diagram of all the forces acting on the box. (4)
- 5.5 Give the reaction force, with direction, of the weight of the box. (2)
- 5.6 Write an equation to express the normal force in terms of F_g . (2)
- 5.7 The angle θ is increased until the box is just about to slide. Calculate the angle θ that will cause the box to start sliding, if the coefficient of static friction between the box and truck is 0,32. Show **all** your calculations. (5)

QUESTION 6

14 MARKS

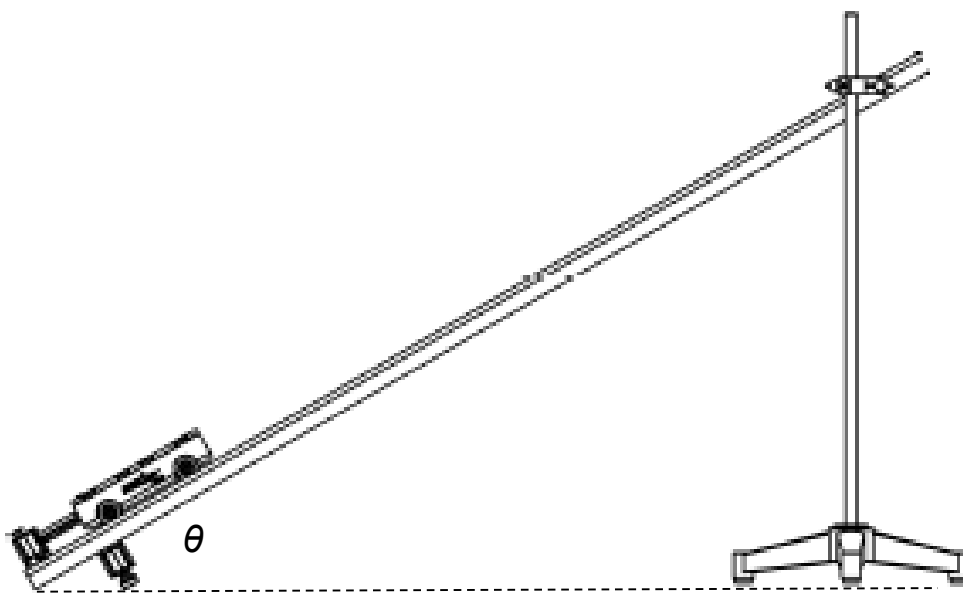
A student was on holiday and saw an escape ramp for trucks. The ramp is used for trucks in an emergency that need to come to a stop when travelling downhill.

The student wonders if there is a relationship between the speed of the truck entering the ramp and the length of the ramp needed for the truck to stop.



6.1 Write a hypothesis for the investigation. (3)

The investigation is recreated in the school laboratory using a Smart Cart, Chromebook and other equipment found in the school laboratory.



6.2 How could the student vary the initial velocity of the cart? (2)

Data is collected and placed in the table below.

velocity of cart (m.s^{-1})	Average displacement (m)	Velocity ² (unit)
0,5	0,07	0,25
1,0	0,29	1,00
1,2	0,42	1,44
1,6	0,75	2,56
1,8	0,95	Question 6.3

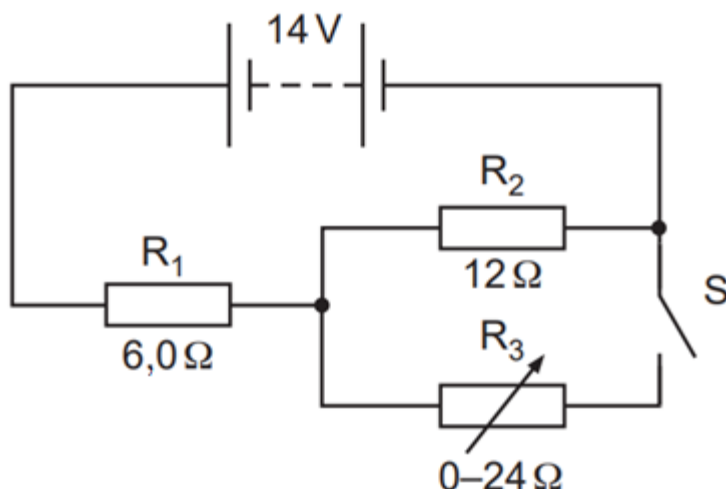
- 6.3 Calculate the missing value in the table. (1)
- 6.4 Plot a graph of velocity² (on the y-axis) vs average displacement (on the x-axis) on the graph paper provided **on the answer sheet**. (6)
- 6.5 Draw a second line on the graph paper provided **on the answer sheet** to show the effect of decreasing the angle θ . Label this line 6.5. (2)

PLEASE TURN OVER

QUESTION 7

26 MARKS

A circuit consists of a battery with an emf of 14 V connected to a resistor network, as shown below. Ignore internal resistance of the battery and connecting wires.



R_1 and R_2 are fixed resistors of resistances $6,0\,\Omega$ and $12\,\Omega$ respectively. R_3 is a variable resistor (rheostat).

7.1 Define *emf*. (2)

7.2 Calculate the resistance of the circuit when the switch is **open**. (2)

Switch S is **closed**.

7.3 What is a short circuit? (1)

7.4 Calculate the current in the battery when the resistance of R_3 is set:

7.4.1 at zero, (4)

7.4.2 at $24\,\Omega$. (4)

7.5 R_3 is set to $24\,\Omega$.

7.5.1 Calculate the total power produced by the battery. (2)

7.5.2 Calculate the voltage drop of R_1 . (3)

7.6 Sketch a graph of voltage of R_1 versus Resistance of R_3 , as R_3 resistance changes from 0 to $24\,\Omega$. (3)

7.7 The rheostat is now removed.

7.7.1 Will the power dissipated in R_1 increase, decrease or remain the same? (1)

7.7.2 Justify your answer in Question 7.7.1 making reference to **two** relevant formulae. (4)

END OF EXAMINATION

EXAMINATION DATA SHEET FOR THE PHYSICAL SCIENCES (PHYSICS)

TABLE 1 PHYSICAL CONSTANTS

NAME	SYMBOL	VALUE
Acceleration due to gravity	g	$9,8 \text{ m}\cdot\text{s}^{-2}$
Speed of light in a vacuum	c	$3,0 \times 10^8 \text{ m}\cdot\text{s}^{-1}$
Universal gravitational constant	G	$6,7 \times 10^{-11} \text{ N}\cdot\text{m}^2\cdot\text{kg}^{-2}$
Coulomb's constant	k	$9,0 \times 10^9 \text{ N}\cdot\text{m}^2\cdot\text{C}^{-2}$
Magnitude of charge on electron	e	$1,6 \times 10^{-19} \text{ C}$
Mass of an electron	m_e	$9,1 \times 10^{-31} \text{ kg}$
Planck's constant	h	$6,6 \times 10^{-34} \text{ J}\cdot\text{s}$
1 electron volt	eV	$1,6 \times 10^{-19} \text{ J}$

TABLE 2 PHYSICS FORMULAE

MOTION

$v = u + at$ or $v_f = v_i + a\Delta t$	$s = \left(\frac{v+u}{2}\right)t$ or $\Delta x = \left(\frac{v_f+v_i}{2}\right)\Delta t$
$v^2 = u^2 + 2as$ or $v_f^2 = v_i^2 + 2a\Delta x$	$s = ut + \frac{1}{2}at^2$ or $\Delta x = v_i\Delta t + \frac{1}{2}a(\Delta t)^2$

FORCE AND MOMENTUM

$F_{net} = ma$	$F_{net} = \frac{\Delta p}{\Delta t}$ or $F_{net}\Delta t = m\Delta v$	$J = \Delta p = mv - mu$ or $J = \Delta p = mv_f - mv_i$
$p = mv$	$F_g = mg$	$F_{fs}^{\max} = \mu_s F_N$ $F_{fk} = \mu_k F_N$

WORK, ENERGY AND POWER

$W = Fs$ or $W = F\Delta x$ or $W = F\Delta x \cos\theta$	$P = \frac{W}{t}$	$P = Fv$
$E_p = mgh$	$E_k = \frac{1}{2}mv^2$	$W_{net} = \Delta E_K$
efficiency = $\frac{\text{power}_{\text{out}}}{\text{power}_{\text{in}}} \times 100$		

GRAVITATIONAL AND ELECTRIC FIELDS

$F = G \frac{m_1 m_2}{r^2}$	$g = \frac{F}{m}$	$g = G \frac{M}{r^2}$
$F = k \frac{q_1 q_2}{r^2}$	$E = \frac{F}{q}$	$E = \frac{kQ}{r^2}$

ELECTRIC CIRCUITS

$I = \frac{q}{t}$	$V = \frac{W}{q}$
$R = \frac{V}{I}$	$emf = I(R_{ext} + r)$ or $emf = V_{load} + V_{internal\ resistance}$
$R_s = R_1 + R_2 + \dots$	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$
$P = \frac{W}{t} \quad \text{or} \quad W = Pt$	
$W = VIt \quad \text{or} \quad W = I^2 R t \quad \text{or} \quad W = \frac{V^2}{R} t$	
$P = VI \quad \text{or} \quad P = I^2 R \quad \text{or} \quad P = \frac{V^2}{R}$	

ELECTRODYNAMICS

$\Phi = BA \cos \theta$	$emf = -\frac{N \Delta \Phi}{\Delta t}$	$F = IB \ell \sin \theta$
$V_p I_p = V_s I_s$	$\frac{N_s}{N_p} = \frac{V_s}{V_p}$	

PHOTONS AND ELECTRONS

$c = f \lambda$		$E = hf$		or	$E = \frac{hc}{\lambda}$
$E = W_0 + E_{K(\max)}$	$W_0 = hf_0$		$E_{K(\max)} = \frac{1}{2}mv_{\max}^2$		